6-2. Multiple regression: Example 6.1

Fit a linear regression model to the data below.

y	x_1	x_2
3.5	3.1	30
3.2	3.4	25
3.0	3.0	20
2.9	3.2	30
4.0	3.9	40
2.5	2.8	25
2.3	2.2	30

The model is $\boldsymbol{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where

$$\boldsymbol{Y} = \begin{pmatrix} 3.5 \\ 3.2 \\ 3.0 \\ 2.9 \\ 4.0 \\ 2.5 \\ 2.3 \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & 3.1 & 30 \\ 1 & 3.4 & 25 \\ 1 & 3.0 & 20 \\ 1 & 3.2 & 30 \\ 1 & 3.9 & 40 \\ 1 & 2.8 & 25 \\ 1 & 2.2 & 30 \end{pmatrix}$$

OLS estimator is $\hat{\boldsymbol{\beta}} = \left(X^T X\right)^{-1} X^T \boldsymbol{Y}$

To calculate this for our example, can use R

First, input the design matrix.

```
> X = matrix(c(1,1,1,1,1,1,1,3.1,3.4,3.0,3.2,
3.9,2.8,2.2,30,25,20,30,40,25,30),7,3)
```

Then compute $\left(X^T X\right)^{-1}$:

```
> XT = t(X)
> XTX = XT %*% X
> XTXinverse = solve(XTX)
```

Also need to compute $X^T \boldsymbol{Y}$:

```
> Y = matrix(c(3.5,3.2,3.0,2.9,4.0,2.5,2.3),7,1)
> XTY = XT %*% Y
```

Finally, compute $\hat{\boldsymbol{\beta}} = \left(X^T X\right)^{-1} X^T \boldsymbol{Y}$:

```
> betahat = XTXinverse %*% XTY
> betahat
             [,1]
[1,] -0.21381852
[2,]  0.89843390
[3,]  0.01745279
```

So the fitted model is

$$\hat{Y} \quad = \quad -0.2138 + 0.8984 x_1 + 0.01745 x_2$$

Can produce the result directly using the `lm` command in `R`

First, copy the appropriate columns of the matrix X into vectors x_1, x_2

```
> x1 = X[,2]
> x2 = X[,3]
```

Then fit the regression model and print a summary.

```
> regression = lm(Y ~ x1 + x2)
> summary(regression)

Call:
lm(formula = Y ~ x1 + x2)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x1           0.89843    0.25300   3.551   0.0238 *
x2           0.01745    0.02116   0.825   0.4558
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

May also want to examine the hat matrix $H = X\left(X^T X\right)^{-1} X^T$

```
> H = X %*% XTXinverse %*% XT
> H
           [,1]       [,2]        [,3]      [,4]        [,5]        [,6]        [,7]
[1,] 0.15269461  0.1077844  0.08383234 0.1497006  0.20059880  0.12574850  0.17964072
[2,] 0.10778443  0.3498388  0.35329341 0.1418701  0.10087517  0.14532474 -0.19898664
[3,] 0.08383234  0.3532934  0.49700599 0.1017964 -0.20359281  0.24550898 -0.07784431
[4,] 0.14970060  0.1418701  0.10179641 0.1543068  0.22984800  0.11423307  0.10824505
[5,] 0.20059880  0.1008752 -0.20359281 0.2298480  0.80953478 -0.07461999 -0.06264394
[6,] 0.12574850  0.1453247  0.24550898 0.1142331 -0.07461999  0.21441732  0.22938738
[7,] 0.17964072 -0.1989866 -0.07784431 0.1082450 -0.06264394  0.22938738  0.82220175
```

Can also find the fitted values $\hat{\boldsymbol{Y}} = H\boldsymbol{Y}$

```
> Yhat = H %*% Y
> Yhat
          [,1]
[1,] 3.094910
[2,] 3.277176
[3,] 2.830539
[4,] 3.184754
[5,] 3.988185
[6,] 2.738116
[7,] 2.286320
```

Again, these may be found from the output of the `lm` command (previously stored with the name `regression`)

```
> fitted.values(regression)
       1        2        3        4        5        6        7
3.094910 3.277176 2.830539 3.184754 3.988185 2.738116 2.286320
```

And can examine residuals $\boldsymbol{e} = \boldsymbol{Y} - \hat{\boldsymbol{Y}}$

```
> Y-Yhat
             [,1]
[1,]  0.40508982
[2,] -0.07717642
[3,]  0.16946108
[4,] -0.28475357
[5,]  0.01181483
[6,] -0.23811608
[7,]  0.01368033
```

Or from output of `lm`

```
> residuals(regression)
          1           2           3           4           5           6           7
 0.40508982 -0.07717642  0.16946108 -0.28475357  0.01181483 -0.23811608  0.01368033
```